

ASSEMBLY LINE BALANCING AS GENERALIZED BIN PACKING *

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Bin packing heuristics are generalized and adapted to solve the assembly line balancing problem. Worst-case analysis is provided. The results are compared to those for a resource constrained scheduling problem considered by Garey, Graham, Johnson and Yao.

Assembly line balancing, bin packing, heuristics

In an assembly line balancing (ALB) problem [3,7], we are given a finite set $L = \{x_1, x_2, \dots, x_n\}$ of tasks, a partial order α (known as a precedence relation) defined on L , and a desired rate of production $1/C$ (C is known as the cycle time). Let $t(x)$ denote the size of task x . We wish to find an ordered partition $P = \langle B_1, B_2, \dots, B_k \rangle$ of L into nonempty sets (known as work stations) such that:

- (i) for each i , the work content at station i is not greater than the cycle time; that is, $\sum_{x \in B_i} t(x) \leq C$;
- (ii) if $x, y \in L$, $x < y$ and $x \in B_i$, $y \in B_j$, then $i \leq j$;
- (iii) k is minimized subject to (i) and (ii).

If (ii) is dropped completely (i.e. no precedence constraint, $\alpha = \emptyset$) the problem is the familiar bin packing problem [1,5]. If the condition $i \leq j$ in (ii) is replaced by $i < j$, then the problem becomes a single resource constrained scheduling (RCS) problem that has been considered in [1].

Two popular bin packing heuristics [5] FIRST FIT (FF) and FIRST FIT DECREASING (FFD) have been generalized and applied to RCS. The worst-case behavior of the generalized bin packing heuristics are also provided in [1]. In this paper we analyze the worst-case behavior of the corresponding generalized bin packing heuristics in the ALB and compare them to the results for RCS.

In FF, elements are packed successively into the first bin in which it fits. FFD is the same as FF except the elements are first sorted in decreasing order of their

size. The generalized FF algorithm works as follows. Initially update the set of available tasks (i.e. tasks whose immediate predecessors have been assigned). Assign the available tasks successively into the first station in which both the precedence and capacity constraints are satisfied. After the assignment of all available tasks, the next set of available tasks is updated and the assignment proceeds. The updating and assignment processes continue until all tasks have been assigned.

We shall show that the generalized FF algorithm can be implemented in time $O(n \log n + e)$, and requires $O(n + e)$ computer storage, where n is the total number of tasks and e is the total number of edges of the ALB precedence graph. As in any sequencing and scheduling problem, precedence relations can be represented by a directed graph. The edges $(a_1, b_1), (a_2, b_2), \dots, (a_e, b_e)$ of the ALB precedence graph denote that task a_i is an immediate predecessor of task b_i , i.e. $a_i \alpha b_i$, $i = 1, 2, \dots, e$. This input structure allows an ALB graph to have 'isolated' nodes. (A node x is said to be isolated if x does not have a follower nor a predecessor.) The computer storage required is $O(n + e)$. Let P_x be the number of unassigned immediate predecessors of task x . The set of available tasks is determined by $\{x | P_x = 0\}$. After an assignment of a task x into a station, the number of unassigned immediate predecessors of each immediate follower y of x (i.e. (x, y) is an input edge) is decreased by 1. If $P_y = 0$, then y is a task in the next set of available tasks. Let $\text{START}(x)$ be the lowest indexed station such that all the immediate predecessors of the task x have been assigned to stations $1, 2, \dots, \text{START}(x)$. That is, task x must be assigned to a station with index

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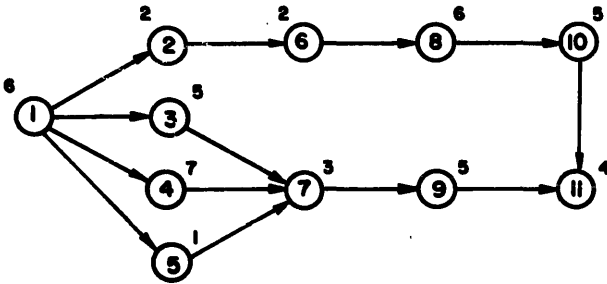


Fig. 1. Jackson's [4] 11-element problem.

greater than or equal to $\text{START}(x)$. The vector START can be computed as follows. Initially, for each task x , put $\text{START}(x) \leftarrow 1$. After an assignment of task x into a station B_x , update $\text{START}(y) \leftarrow \text{MAX}(\text{START}(y), B_x)$, for each immediate follower y of x . Thus the generalized FF algorithm can be implemented in time $O(n \log n + e)$, and requires $O(n + e)$ computer storage.

The generalized FFD algorithm is the same as the generalized FF algorithm except that at each stage the set of available tasks are sorted in decreasing order of their size.

As in RCS [1], we have the LEVEL algorithm (FFL) for the ALB. FFL records each set of available tasks into the form $\langle v_1, v_2, \dots, v_m \rangle$, where $\text{level}(v_1) > \text{level}(v_2) > \dots > \text{level}(v_m)$. The level of task x is the length of the longest chain which has x as its head. An example to illustrate these algorithms is given in Fig. 1.

Initially, only task 1 is available, hence it is assigned to station 1. The next set of available tasks is $\{2, 3, 4, 5\}$. In generalized FF the order of the tasks to be considered for assigning is in the given order. Task 2 is assigned to station 1, task 3 is assigned to the 2nd station since $t(1) + t(2) + t(3) > C$, etc. In generalized FFD, the available tasks are sorted in decreasing task size order so the next ordered set of available tasks is $\langle 4, 3, 2, 5 \rangle$. In generalized FFL, the next ordered set of available tasks is $\langle 2, 3, 4, 5 \rangle$. The chain with task 2 as

head is 2-6-8-10-11 which is of length 4, i.e. $\text{level}(2) = 4$, $\text{level}(x) = 3$, for $x = 3, 4, 5$. In FFD and FFL assign all the tasks in the sorted order and then update the set of available tasks.

The generalized FFD of RCS [1] has worst-case ratio lying in the range $(2.691, 2.7)$. For generalized FF and FFL of RCS the bounds are $\frac{1}{2}(\text{OPT} + 3)$ and 2.7 respectively, where OPT is the optimal number of bins [1]. We shall show that the worst-case ratios of the corresponding generalized FF, FFD and FFL of ALB are 2. Let $H(L)$ be the number of stations obtained by applying heuristic H to an ALB problem L , and $\text{OPT}(L)$ be the optimal number of stations. Since every pair of consecutive work stations obtained by the generalized H ($= \text{FF}, \text{FFD}$ or FFL) has sum of tasks exceeding C , the total work content (defined as the sum of all tasks in L) $W > \frac{1}{2}H(L) \times C$, if $H(L)$ is even; and $W > (H(L) - 1) \times \frac{1}{2}C$, if $H(L)$ is odd. Let $[W]$ be the smallest integer $\geq W$. Then $[W] \leq C \times \text{OPT}(L)$ which implies $H(L) \leq 2 \times \text{OPT}(L) - 2$, if $H(L)$ is even and $H(L) \leq 2 \times \text{OPT}(L) - 1$, otherwise. Thus $H(L) \leq 2 \times \text{OPT}(L) - 1$, for any L . We shall next show how to construct lists L for which $H(L)/\text{OPT}(L)$ is arbitrarily close to 2, showing that the upper bound 2 is essentially the best possible.

For any given cycle time $C > 0$ and integer $k \geq 1$ construct the ALB problem in Fig. 2, in which the number of elements of size 3δ is $2k - 1$ and $0 < \delta \leq \frac{1}{6}C/k$. The optimal solution is $k + 1$ and is given by

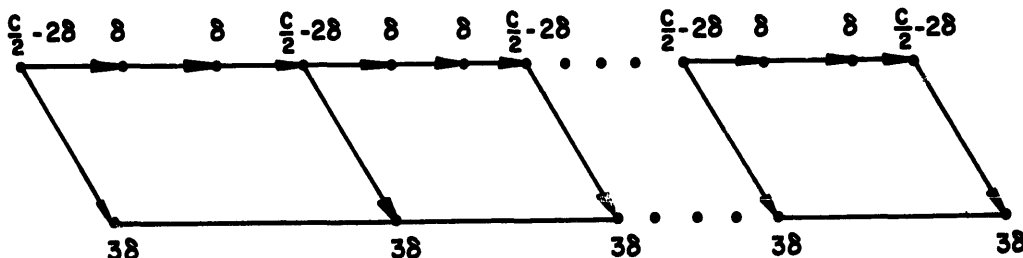
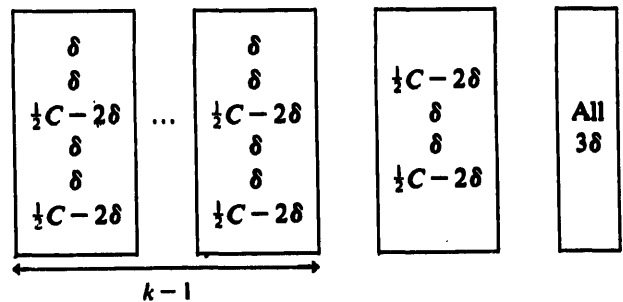
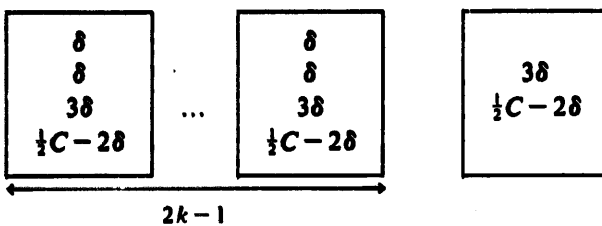


Fig. 2.

Table 1

	NF	FF	FFL	FFD	BF	BFL	BFD	EPSI	MFFD
Bin packing: $\alpha = \emptyset$	2	1.7	1.7	$\frac{11}{9}$	1.7	1.7	$\frac{11}{9}$	$\frac{11}{9} - \epsilon$	$[\frac{70}{60}, \frac{6}{5}]$
ALB: $\alpha = \emptyset$	2	2	2	2	2	2	2	2	2
Time complexity	T_1	T_2	T_2	T_2	T_2	T_2	T_2	T_3	T_2

and for FF, FFD or FFL, the solution is $2k$ and is



Other well-known heuristics, NEXT FIT (NF), BEST FIT (BF) [5], Yao's EPSI algorithm [10], as well as Garey and Johnson's MFFD [2] can also be generalized to the ALB problem in the same manner as FF. The worst-case ratios and the time complexities of these generalized heuristics are summarized in Table 1, where $\epsilon = 10^{-9}$, $T_1 = O(n + e)$, $T_2 = O(n \log n + e)$, $T_3 = O(n^{10} \log n)$.

We have tested several of these heuristics on problems from the literature, as well as randomly generated problems and have found very encouraging empirical results [6]. In additions, they have been incorporated into an enumeration scheme for two versions of the assembly line balancing problem [8,9].

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